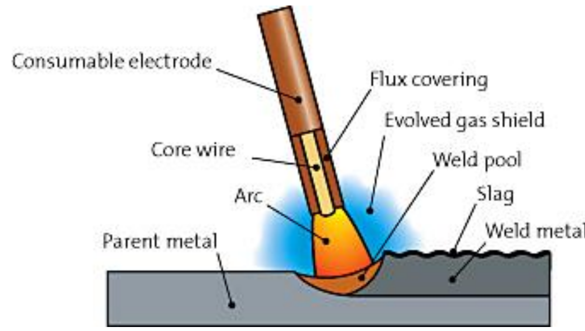
6.2 Welded Connections

- The most common welding method is ***arc welding*** which consists of generating high temperatures from an electric arc to melt the metal at the joint between two parts.

- Welded connections require more control in the fabrication process compared to bolted connections.



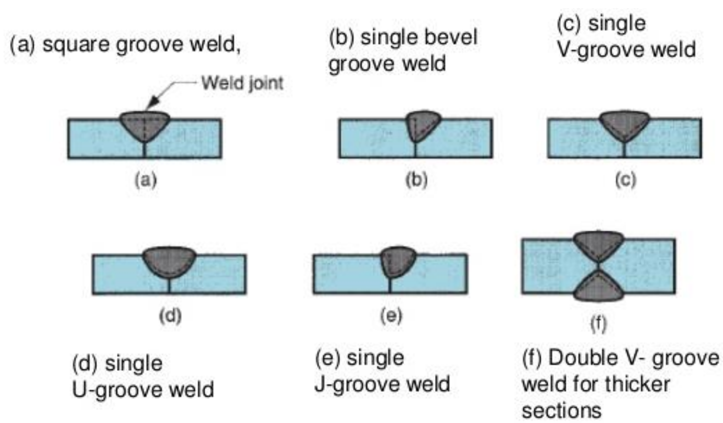
- Inspection of the weld requires significant expertise and experience (high labour cost)

- Welded connections can be more economical than bolted connections if made in the fabricator's shop instead of the field.

6.2.1 Weld Types

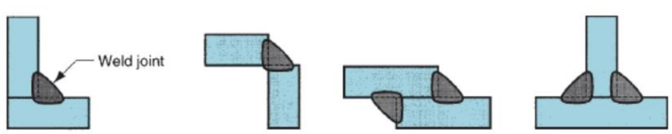
i. Groove weld

- Can be either single groove weld (partial penetration) or double grove weld (complete penetration).
- Double groove weld can restore the original base metal strength if the weld metal has at least the same strength as the base metal.
- The single groove weld is less effective than the double groove weld.



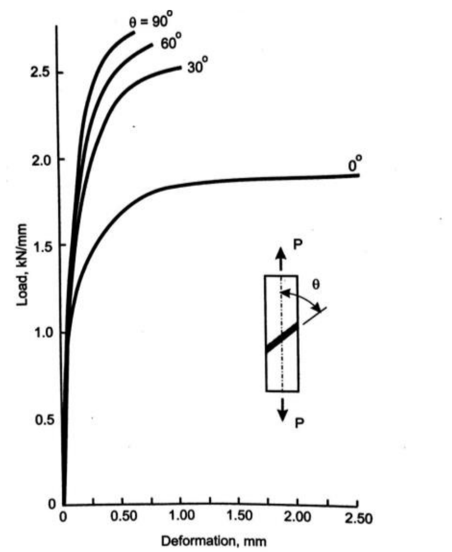
ii. Fillet weld

- Easier to make than the groove weld, but it is less effective.



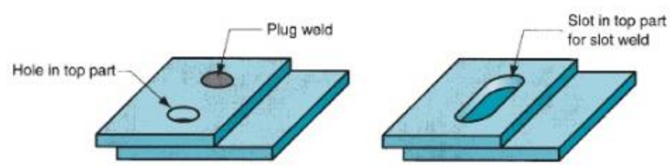
Fillet weld on corner joint Fillet weld on lap joint Fillet weld on T-joint

- The strength and ductility of the fillet weld depend on the angle of inclination between the weld and the load.



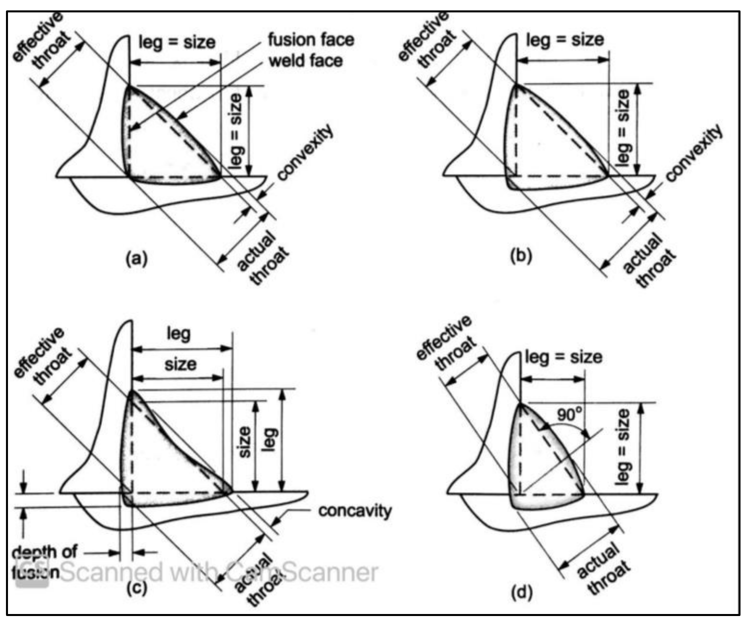
iii. Plug (slug) weld

Are used less frequently in structural engineering.

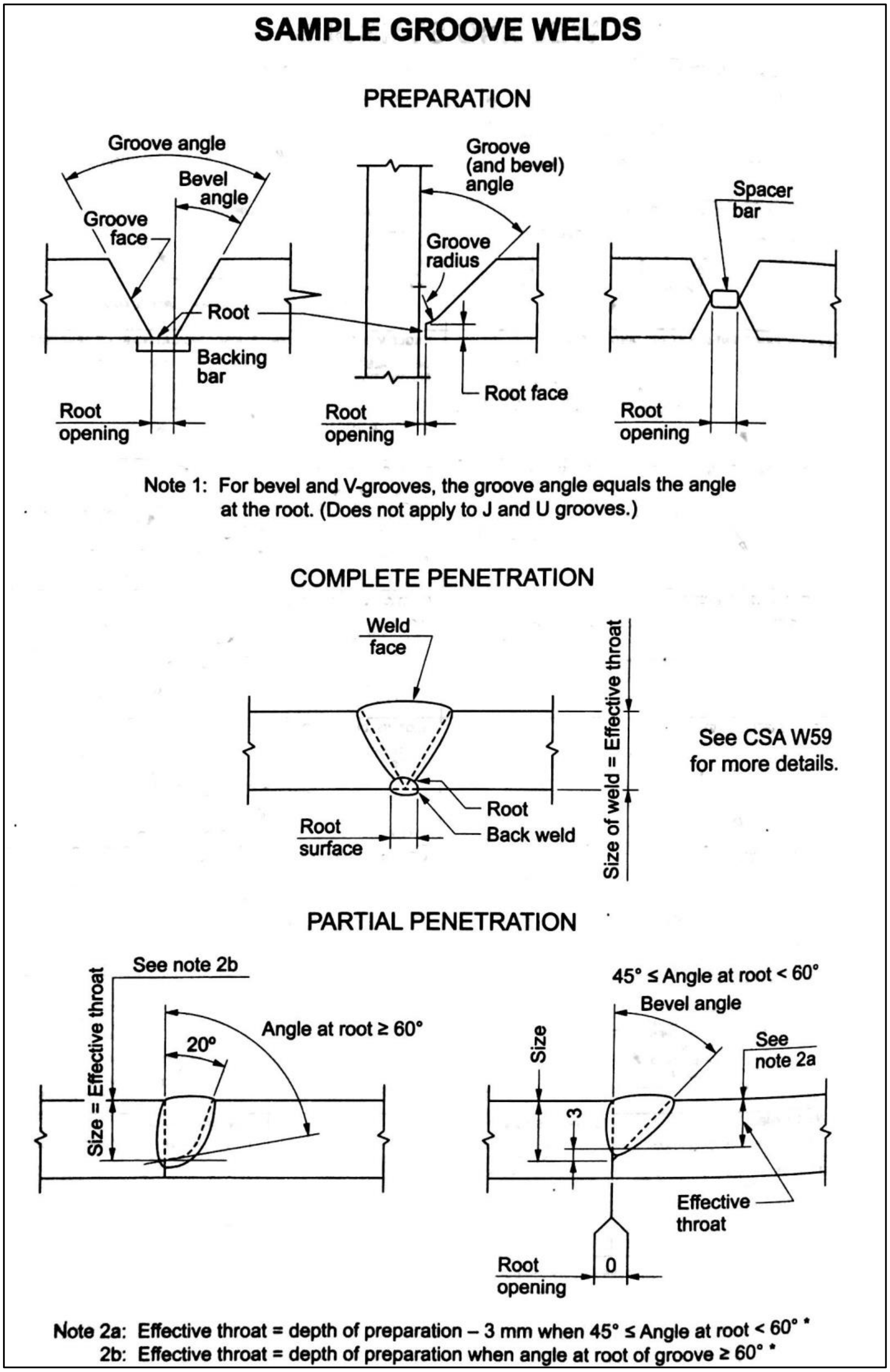


<u>Weld Terminology and Symbols</u>

1. Fillet weld terminology



2. Groove weld terminology



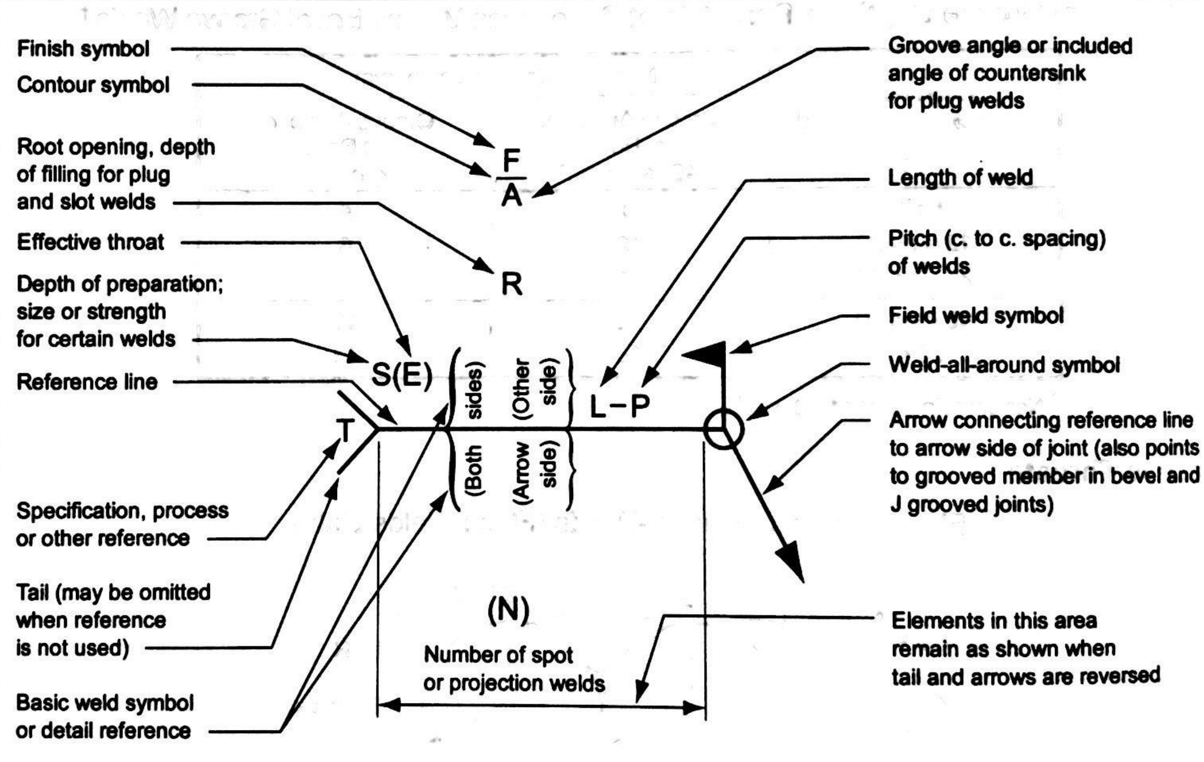
BASIC WELD SYMBOLS

BACK	FILLET	PLUG OR SLOT	GROOVE OR BUTT						
			SQUARE	V	BEVEL	U	J	FLARE V	FLARE BEVEL

SUPPLEMENTARY WELD SYMBOLS

WELD ALL AROUND	* FIELD WELD	CONTOUR		
		FLUSH	CONVEX	CONCAVE
	OR			

STANDARD LOCATION OF ELEMENTS OF A WELDING SYMBOL



Notes:

Size, weld symbol, length of weld and spacing must read in that order from left to right along the reference line. Neither orientation of reference line nor location of the arrow alter this rule.

The perpendicular leg of , , , weld symbols must be at left.

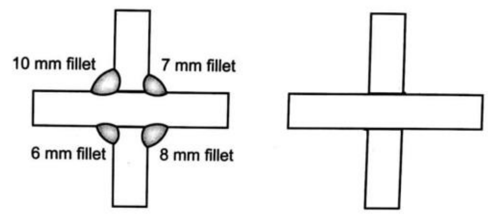
Size and spacing of fillet welds must be shown on both the Arrow Side and the Other Side Symbol.

Symbols apply between abrupt changes in direction of welding unless governed by the "all around" symbol or otherwise dimensioned.

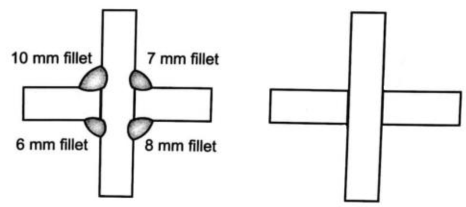
These symbols do not explicitly provide for the case that frequently occurs in structural work, where duplicate material (such as stiffeners) occurs on the far side of a web or gusset plate. The fabricating industry has adopted this convention: when the billing of the detail material discloses the identity of far side with near side, the welding shown for the near side shall also be duplicated on the far side.

Example 1

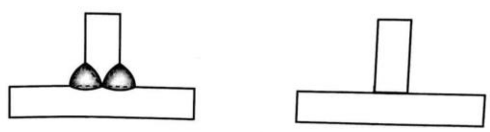
Draw the weld symbol for each connection.



(a) Fillet welds in a cruciform joint (horizontal joints)



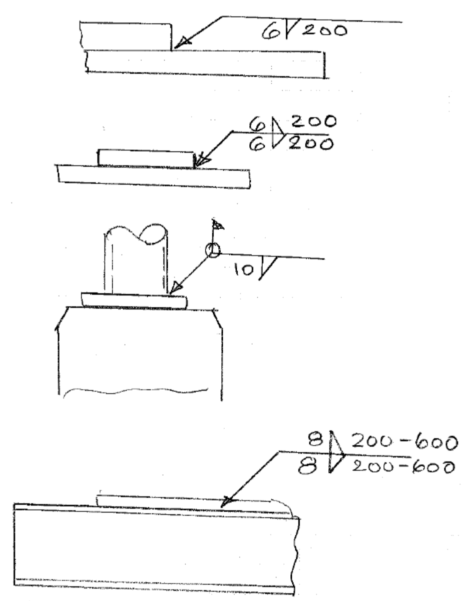
(b) Fillet welds in a cruciform joint (vertical joints)



(c) Groove weld in a T-joint with fillet weld reinforcement

Example 2

What does each weld symbol mean?



6.2.2 Weld Design According to CSA S16

For welded connections the code requires checking the shear resistance of the weld itself and the base metal adjacent to the weld.

13.13 Welds

13.13.1 General

The resistance factor, ϕ_w, for welded connections shall be taken as 0.67.

Note: *See Table 4 for matching electrode classifications for CSA G40.21 steels.*

13.13.2 Shear

13.13.2.1 Complete and partial joint penetration groove welds, and plug and slot welds

The factored shear resistance shall be taken as the lesser of

a) for the base metal:
$V_r = 0.67\phi_w A_m F_u$

b) for the weld metal:
$V_r = 0.67\phi_w A_w X_u$

where
A_m = shear area of effective fusion face
A_w = area of effective weld throat, plug, or slot

13.13.2.2 Fillet welds

The factored resistance for direct shear and tension- or compression-induced shear shall be taken as

$$V_r = 0.67\phi_w A_w X_u (1.00 + 0.50 \sin^{1.5}\theta) M_w$$

where
θ = angle, in degrees, of axis of weld segment with respect to the line of action of applied force (e.g., 0° for a longitudinal weld and 90° for a transverse weld)
M_w = strength reduction factor for multi-orientation fillet welds. For joints with a single weld orientation, M_w = 1.0; for joints with multiple weld orientations, for each segment

$$M_w = \frac{0.85 + \theta_1 / 600}{0.85 + \theta_2 / 600}$$

where
θ_1 = orientation of the weld segment under consideration
θ_2 = orientation of the weld segment in the joint that is nearest to 90°

Weld returns that are not accounted for in the joint capacity need not be considered a weld segment in the context of this Clause.

Where over-matched electrodes are used, the base metal capacity at the fusion face shall also be checked and may be considered to have the following strength:

$$V_r = 0.67\phi_w A_m F_u$$

13.13.2.3 Flare bevel groove welds for open-web steel joists

The factored resistance for direct shear and tension- or compression-induced shear shall be taken as

$V_r = 0.67\phi_w A_w F_u$

where

A_w = $0.50 w_f L$ (or as established by procedure qualification tests)

where

w_f = width of flare bevel groove weld face

F_u = least ultimate tensile strength of the components in the joint

Table 4
Matching electrode ultimate tensile strengths for CSA G40.21 steels
(See Clause 13.13.1.)

Matching electrode ultimate tensile strength* MPa	G40.21 Grades, MPa						
	260	300	350	380	400	480	700
430	X	X†					
490	X	X	X‡	X			
550					X‡		
620						X	
820							X

** The electrode ultimate tensile strength is ten times the first two digits of the electrode classification in CSA W48.*
† For HSS only.
‡ For unpainted applications using "A" or "AT" steels where the deposited weld metal is to have atmospheric corrosion resistance or colour characteristics, or both, similar to the base metal, the requirements of Clauses 5.2.1.4 and 5.2.1.5 of CSA W59 shall apply.

13.13.3 Tension normal to axis of weld

13.13.3.1 Complete joint penetration groove weld made with matching electrodes

The factored tensile resistance shall be taken as that of the base metal.

13.13.3.2 Partial joint penetration groove weld made with matching electrodes

The factored tensile resistance shall be taken as

$$T_r = \phi_w A_n F_u \le \phi A_g F_y$$

where
A_n = nominal area of fusion face normal to the tensile force

When overall ductile behaviour is desired (member yielding before weld fracture), the following shall apply:

$$A_n F_u > A_g F_y$$

13.13.3.3 Partial joint penetration groove weld combined with a fillet weld, made with matching electrodes

The factored tensile resistance shall be taken as

$$T_r = \phi_w \sqrt{(A_n F_u)^2 + (A_w X_u)^2} \le \phi A_g F_y$$

where
A_g = gross area of the components of the tension member connected by the welds

13.13.4 Compression normal to axis of weld

13.13.4.1 Complete and partial joint penetration groove welds made with matching electrodes

The compressive resistance shall be taken as that of the effective area of base metal in the joint. For partial joint penetration groove welds, the effective area in compression shall be taken as the nominal area of the fusion face normal to the compression plus the area of the base metal fitted in contact bearing (see Clause 28.5).

13.13.4.2 Cross-sectional properties of continuous longitudinal welds

Continuous longitudinal welds made with matching electrodes may be considered as contributing to the cross-sectional properties A, S, Z, and I of the cross-section.

13.13.4.3 Welds for hollow structural sections

The provisions of Annex L of CSA W59 may be used for hollow structural sections.

6.2.3 Weld Detailing

Minimum Size

The minimum fillet size as measured should be as shown in the table, unless a larger size is required to meet the calculated resistance. This minimum size requirement need not apply when:

a) welding attachments to members without calculated stress or
b) welding procedures have been established to prevent cracking in accordance with W59-13.

Material thickness t (mm)	Minimum fillet size (mm)
$t \le 6$	3*
$6 < t \le 12$	5
$12 < t \le 20$	6
$20 < t$	8

* *For cyclically-loaded structures min. size = 5 mm*

For consumables with a hydrogen content conforming to the H8 requirement or lower, t is the thickness of the thinner part joined. Otherwise, t is the thickness of the thicker part joined; the weld size, however, need not exceed the thickness of the thinner part provided particular care is taken to provide sufficient heat input to ensure weld soundness.

The minimum effective length of a fillet weld should be 38 mm or 4 times the size of the fillet, whichever is larger. Where the geometry of the joint makes it impossible to deposit the minimum effective length, the effective fillet size shall be 0.25 times its effective length.

Maximum Size

The maximum fillet weld size, D_{max}, recommended by good practice along a sheared edge is:

$D_{max} = t$ when $t < 6$ mm
$D_{max} = t - 2$ when $t \ge 6$ mm

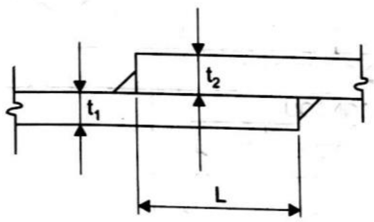
When fillet welds are used in holes or slots, the diameter of the hole or the width of the slot should not be less than the thickness (t) of the member containing it plus 8 mm. The maximum diameter or width shall be t + 12 mm or 2.25 t, whichever is greater.

$D_{max} = t$ when $t < 6$ mm

$D_{max} = t - 2$ when $t \geq 6$ mm

When fillet welds are used in holes or slots, the diameter of the hole or the width of the slot should not be less than the thickness (t) of the member containing it plus 8 mm. The maximum diameter or width shall be $t + 12$ mm or $2.25\,t$, whichever is greater.

Lap Joints



$L_{min} = 5\,t_1 \geq 25$ mm when $t_1 \leq t_2$

$L_{min} = 5\,t_2 \geq 25$ mm when $t_2 < t_1$

Partial Penetration Groove Welds

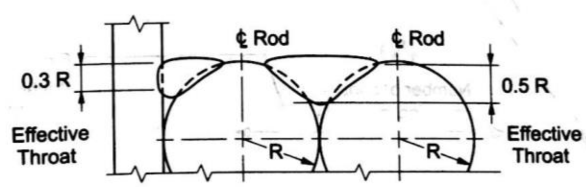
Minimum Groove Depth for Partial Joint Penetration V-, and Bevel Groove Welds †

Thickness, t of Thicker Part Joined (mm)	Minimum Groove Depth, mm	
	Groove Angle, α, at Root $45° \leq \alpha < 60°$	Groove Angle, α, at Root $\alpha \geq 60°$
$t \leq 12$	8	5
$12 < t \leq 20$	10	6
$20 < t \leq 40$	11	8
$40 < t \leq 60$	12	10
$60 < t$	16	12

† Not combined with fillet welds

Effective Throats

Flare Bevel and Flare V-Welds (Flush Welds Only)



Solid or hollow sections with weld filled flush to the curved surface:

Not applicable to flare V-welds using GMAW process except when $R \geq 12$ mm, in which case the effective throat = $0.375R$.

Flare Bevel Groove Weld

When $R > 10$ mm, the effective throat for a joint between a curved and a planar surface shall be $0.3\,R$. When $R \leq 10$ mm, design as a fillet weld unless an effective throat has been previously qualified as a Flare Bevel (See W59 Clause 4.3.1.6.2.2).

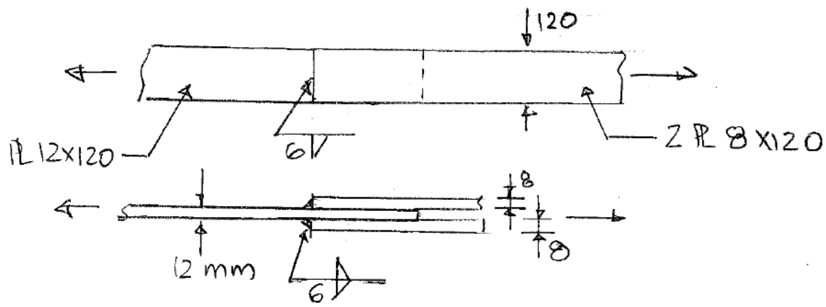
Flare Vee Groove Weld

When $R > 10$ mm, the effective throat for a joint between two curved surfaces shall be $0.5\,R$.

Example 4

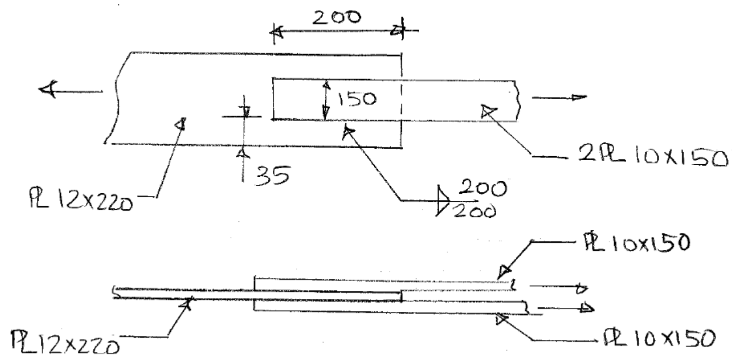
Calculate the capacity of the following connection assuming:

- all plates are made from Grade 300W steel (F_y = 300 MPa and F_u = 440 MPa).
- 6 mm transverse fillet weld is used for the connection.



Example 5

Design the welded connection shown below such that the factored tension resistance of the plates can be reached. Assume Grade 350W (F_y = 350 MPa and F_u = 450 MPa) for all plates.



References:

1. *Handbook of Steel Construction* by the Canadian Institute of Steel Construction
2. *Limit States Design in Structural Steel* by Kulak Grondin